Problems:

1 **Two bodies A and B of equal masses are suspended from the springs of spring constants k_1 and k_2 respectively. If the bodies oscillate such that their maximum velocities are equal. Find the ratio of their amplitudes.**

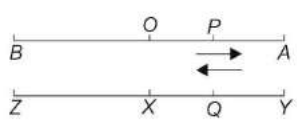
2 **An object of mass 0.2 kg executes SHM along the X -axis with frequency of $\left(\frac{25}{\pi}\right)$ Hz. At the point $= 0.04$ mm, the object has KE $= 0.5$ J and PE $= 0.4$ J. Find the amplitude of the motion.**

3 **A mass m attached to a spring oscillates with a period of 2 s . If the mass is increased by 2 kg , the period increases by 1 s . Assume Hooke's law is obeying, then calculate initial mass m.**

4 A particle is executing SHM of amplitude 2 cm . At extreme position, the force is 4 N . How much force will be at the mid-point between the mean and extreme position?

5 **In case of a SHM, (a) what fraction of total energy is kinetic and what fraction is potential, when displacement is one-half of the amplitude? (b) at what displacement, the kinetic and potential energies are equal?**

6. **Two particles execute SHM of same amplitude and frequency along the same straight line. They cross each other while going in opposite direction each time, their displacement is half of their amplitude. Find the phase difference between them.**



7. The displacement of a particle executing periodic motion is given by

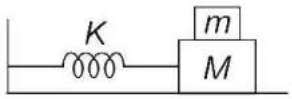
$$Y = 4\cos^2\left(\frac{1}{2}t\right)\sin 1000t$$

Find the independent components of simple harmonic motion of which it is superposition.

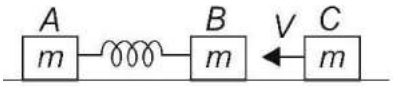
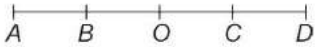
8. A block is resting on a platform which is moving vertically with SHM of period 1.0 s . At what amplitude of motion block and platform will separate?

9. Two simple pendulums of length 1 m and 16 m respectively are both given small displacement in the same direction and at the same instant. They will be again in phase after the shorter pendulum completed n oscillations. Calculate n.

10. A horizontal spring block system (force constant) executes SHM with amplitude A. When the block of mass M is passing through its equilibrium position, an object of mass m is put on it and the two move together. Find the new amplitude and frequency of vibration.



11. Two blocks each of mass m are connected by a spring of stiffness constant k. A third identical block C and mass m moves with velocity V towards blocks A and B. If the collision is elastic, then calculate the maximum compression of the spring and the kinetic energy of the blocks A and B at this compression.



12. A particle is executing simple harmonic motion in a straight line $ABCD$. Its KE at B and C is half the maximum KE . If period of oscillation of particle is T and amplitude a, then find the time taken by the particle to go from O to B.